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Multiple Concussions Increase Odds and Rate of Lower Extremity Injury in National Collegiate Athletic Association Athletes After Return to Play

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Background: Concussion in collegiate athletics is one of the most prevalent sport-related injuries in the United States, with recent studies suggesting persistent deficits in neuromuscular control after a concussion and an associated increase in risk of lower extremity injury.

Purpose: To expand on the relationship between concussion and lower extremity injury by examining the effect of multiple concussions (MC) on rate and odds of future lower extremity injury in collegiate athletes after return to play (RTP) compared with matched controls.

Study Design: Cohort study; Level of evidence, 3.

Methods: From 2001 to 2016, 48 National Collegiate Athletic Association Division I athletes sustaining multiple concussions at a single institution were identified. Athletes with multiple concussions (MC) were matched directly to athletes with a single concussion (SC) and to athletes with no concussion history (NC) by sex, sport, position, and games played. Incidence of, time to, and location of lower extremity injury were recorded for each group after RTP from their first reported concussion until completion of their collegiate career. Logistic regression was used to analyze odds ratios (ORs) for sustaining lower extremity injury, whereas time to injury was summarized by use of Kaplan-Meier curves and log rank test analysis.

Results: The incidence of lower extremity injury after RTP was significantly greater (P = .049) in the MC cohort (36/48, 75%) than in SC athletes (25/48 = 52%) and NC athletes (27/48 = 56%). Similarly, odds of lower extremity injury were significantly greater in the MC cohort than in SC athletes (OR, 3.00; 95% CI, 1.26-7.12; P = .01) and NC athletes (OR, 1.66; 95% CI, 1.07-2.56; P = .02). Time to lower extremity injury was significantly shorter in the MC group compared with matched controls (P = .01). No difference was found in odds of lower extremity injury or time to lower extremity injury between SC and NC athletes.

Conclusion: Collegiate athletes with MC were more likely to sustain a lower extremity injury after RTP in a shorter time frame than were the matched SC and NC athletes. This may suggest the presence of a cohort more susceptible to neuromuscular deficits after concussion or more injury prone due to player behavior, and it may imply the need for more stringent RTP protocols for athletes experiencing MC.

Keywords: head injuries/concussion; general sports trauma; football (American); injury prevention

Concussions are among the most common sport-related injuries in the United States. Most recent data indicate that approximately 3.42 million emergency department visits were attributable to sport-related concussions from 2001 to 2012 and that many more concussions likely go unreported.[3,10,27] Furthermore, within popular collegiate sports such as basketball, football, and soccer, concussions are thought to constitute between 3.6% and 13.1% of all sport-related injuries.[29] Even in sports with lower national participation, concussion continues to be a significant risk. For example, Blumenfeld et al[5] noted in 2016 that 36% of water polo survey respondents had previously experienced a concussion during competitive play. Despite this frequency, concussions continue to be one of the most difficult conditions to diagnose and manage due to the variability in presentation and mysteries that surround the pathophysiological process. Indeed, in recent years, fewer athletes have displayed clinically obvious symptoms, such as loss of consciousness or amnesia, and more athletes have shown occult, nonspecific presentations, such as drowsiness, irritability, or light sensitivity.[11] Although this change in presentation may largely be attributable to the recent paradigm shift in clinician

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awareness of concussions, it underscores the complexity of this condition and its burden on athletic competition.

Recent efforts have been taken to improve player safety and increase detection of concussion in sport. At present, sport concussion is frequently diagnosed and assessed through use of subjective symptom questionnaires and neuropsychological testing. These tests, combined with the athlete's ability to tolerate a gradual return to aerobic activity, ultimately dictate when an athlete is deemed capable of return to play (RTP). However, many studies suggest that current RTP protocols are limited in their ability to effectively predict complete resolution of postconcussive symptoms. Brooks et al[6] found that recently concussed athletes had a 2.48 times greater odds of sustaining a lower extremity injury in the 90 days after RTP. Herman et al[18] found a similar relationship, measuring a 3.39 times greater odds of lower extremity injury in the same 90-day interval. Furthermore, sensitive examination of balance and gait seems to corroborate these findings, as postconcussion athletes have measurable deficits in these metrics long after documented RTP.[12,17,20] These results suggest the need for improvements in modern concussion RTP protocols as well as further study on the pathophysiological process of concussion.

Given these findings, it is logical to assume that special consideration should be given to athletes who experience multiple concussions during their career. This is especially true given that mild traumatic brain injuries potentially increase the risk for further concussions and lasting cognitive deficits.[9,25] Some estimates predict that a single athletic concussion can triple the risk for a second concussion.[8] Furthermore, other studies have demonstrated cumulative and persistent adverse cognitive effects, worsening with each incidence of head trauma.[28] Although data regarding these outcomes are currently debated, with some studies demonstrating no lasting cognitive deficits and other factors related to increased concussion risk, the potential consequences of repeated head trauma remain a significant concern.[2,15] Recent studies on increased injury risk after concussion suggest that athletes who experience multiple concussions throughout their career may have greater deficits than those who experience a single concussive event. As such, the primary aim of this study was to explore this relationship, with the presumption that more concussions would lead to a cumulative increase in risk of lower extremity injury. To do this, we performed a retrospective cohort study of concussed National Collegiate Athletic Association (NCAA) athletes over a 15-year period, examining for lower extremity injuries after the first documented concussion.

METHODS

After obtaining approval from an institutional review board, we retrospectively examined sports injury data for NCAA Division I collegiate athletes from 2001 to 2016 by using Sports Injury Monitoring System software (Flan-Tech Inc) at a single athletic institution. The athletes were then divided into 2 cohorts based on concussion status. Multiple concussion (MC) athletes were defined as any athletes sustaining 2 or more concussions in their collegiate career, whereas single concussion (SC) athletes had 1 documented concussion. All athletes were then followed from their first documented concussion until completion of their collegiate career for lower extremity injuries. Concussions were diagnosed using Concussion in Sport (CIS) consensus criteria by licensed sports medicine trainers and physicians. Athletes were then periodically reassessed for resolution of symptoms and advanced in a stepwise fashion in accordance with published guidelines by CIS criteria. Concussion protocol guidelines did not differ significantly during the 15-year study interval.

The MC athletes were then matched with SC athletes and with healthy participants with no concussion (NC) by sex, sport, position, and games played. Regarding specific matching criteria, matching by sex and sport was exact, whereas games played was allowed to differ by no more than 5 games across the entire trio. In rare circumstances, where games played could not meet this criterion, SC and NC athletes with greater exposure were matched to MC athletes with fewer games played (2/48 matched trios). NC participants were followed from the same concussion RTP date as their matched SC athlete. All matches were members of the same active athletic roster at the time of study. Athletes were excluded if they had incomplete roster, injury, and/or demographic information; if appropriate matches could not be found; and if the athlete had yet to complete his or her collegiate career. Demographic information for all nonmatched characteristics was obtained, including height, weight, body mass index, and age.

After exclusions, only 4 sports remained eligible for the study. Matching within a given sport was varied to best account for differences in risk between different positions. The methods used here are in accordance with matching techniques reported in other published studies.[1,4-6,14] Football players were matched by nonskilled (eg, offensive/defensive linemen) or skilled positions (eg, quarterback, wide receiver, linebacker).[4,6,14] Basketball players were matched by guards/forwards or centers. Water polo players were matched indiscriminately except for goalkeepers.[5]

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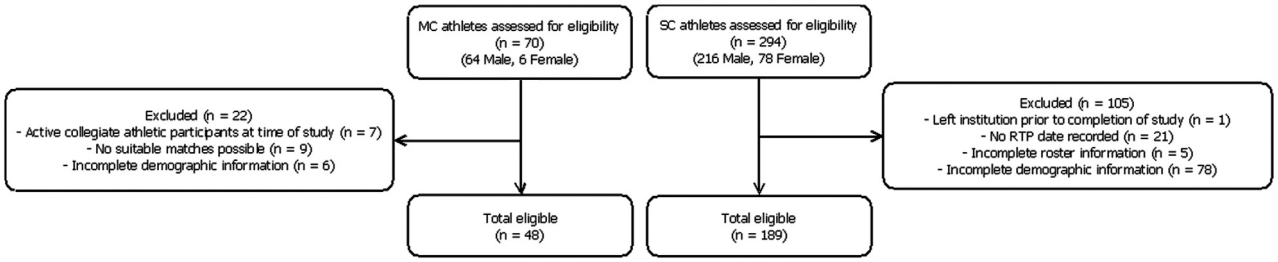

Figure 1. Selection of the eligible cohort. MC, multiple concussions; RTP, return to play; SC, single concussion.

TABLE 1
Patient Characteristics[a]

	Multiple Concussions (n = 48)	Single Concussion (n = 48)	No Concussion (n = 48)	*P* Value
Sex, n[b]				
Male	48	48	48	
Female	0	0	0	
Height, in	74 ± 7.37	73.9 ± 5.4	73.44 ± 5.57	.37
Weight, lb	227.19 ± 93.37	224.08 ± 82.44	221.38 ± 83.77	.51
Body mass index	28.95 ± 8.63	28.78 ± 9.28	29.73 ± 8.68	.80
Age, y	19.27 ± 2.34	19.38 ± 2.55	19.69 ± 2.6	.10
Sport, n				
Football	42	42	42	
Basketball	3	3	3	
Soccer	1	1	1	
Water polo	2	2	2	

[a]Values are expressed as mean ± SD unless otherwise noted.
[b]After exclusions, no female athletes were eligible for study.

Soccer players were matched by forwards/midfielders, defenders, and goalkeepers.[1]

Nonmatched baseline characteristics were compared through use of a combination of chi-square tests and 2-way analysis of variance. Differences in type, location, and nature of lower extremity injury were assessed by chi-square and Fisher exact tests. Binary logistic regression was used to determine the odds ratio (OR) associated with number of concussions and lower extremity injury, whereas Kaplan-Meier and log rank test analysis was used to compare time with injury in each cohort. Survival curves were drawn and censored to 90 days, 1 year, and the graduation date for each respective athlete. Statistical significance was set to $P < .05$ and associated 95% CI. All analysis was performed using RStudio version 1.1.453.

RESULTS

In this study, 70 athletes were noted to have multiple documented concussions, whereas 294 were found to have isolated cases. After exclusions, 48 athletes were deemed eligible for the MC group (34/48 with 2 concussions only, 11/48 with 3 concussions, 3/48 with 4 concussions). From the athletes with SC, 189 athletes were eligible after exclusions, from which a corresponding 48 athletes were chosen for the SC group, based on matching for sex, sport, position, and year of eligibility (Figure 1). The NC group was subsequently selected from the same active roster as the MC and SC cohorts through use of similar criteria. After exclusions, no female athletes were eligible for the study. The final cohort was composed of 48 male athletes per group. The participants represented 4 different sports, with the vast majority of participants derived from football (42/48). Basketball (3/48), water polo (2/48), and soccer (1/48) composed the remainder of the cohort (Table 1).

Among these athletes, 227 lower extremity injuries were identified across the 3 groups. Most of these injuries were found in the MC cohort (114/227 [50%]), whereas fewer were found in the SC and NC cohorts (63/227 [28%] and 50/227 [22%], respectively). A higher proportion of MC athletes experienced at least 1 lower extremity injury after concussion compared with matched controls (MC, 36/48 [75%]; SC, 25/48 [52%]; NC, 27/48 [56%]; $P = .049$). Furthermore, when type, location, and nature of lower extremity injury were examined, athletes with MC were more likely to sustain an injury to the foot ($P = .001$) than were the matched SC and NC athletes (Table 2).

In the 90 days after RTP from concussion, the 3 groups had no difference in odds of sustaining a lower extremity injury. However, at 1 year after RTP, the odds of sustaining a lower extremity injury in the MC group were

TABLE 2
Type, Location, and Chronicity of Lower Extremity (LE) Injury[a]

	Multiple Concussions	Single Concussion	No Concussion	*P* Value
Total LE injuries	114	63	50	
LE injury type				
Contusion	13	10	5	.59
Dislocation	0	2	0	.12
Fracture	2	1	1	.99
Sprain/strain	84	46	42	.30
Stress	3	0	0	.44
Unspecified	12	4	2	.31
LE injury location				
Hip	5	6	2	.31
Thigh	26	17	12	.82
Knee	27	22	20	.07
Shin/calf	8	1	2	.26
Ankle	25	13	13	.78
Foot	23	4	1	**.001**
Chronicity of LE injury				
Acute	101	61	48	.78
Subacute	3	0	1	.55
Chronic/overuse	10	2	1	.14

[a]Values are expressed as numbers of patients. Boldface indicates significance.

TABLE 3
Odds of Lower Extremity Injury[a]

	90 Days		1 Year		Total Study Period	
	OR (95% CI)	*P* Value	OR (95% CI)	*P* Value	OR (95% CI)	*P* Value
SC vs NC	1.00 (0.27-3.70)	≥.999	1.10 (0.47-2.59)	.83	0.92 (0.41-2.05)	.84
MC vs NC	1.50 (0.84-2.69)	.17	1.83 (1.20-2.79)	.005	1.66 (1.07-2.56)	.02
MC vs SC	2.26 (0.71-7.21)	.17	3.06 (1.33-7.03)	.01	3.00 (1.26-7.12)	.01

[a]Total study period varied between athletes based upon date of graduation for the MC athlete. MC, multiple concussions; NC, no concussion; OR, odds ratio; SC, single concussion.

significantly greater than in matched SC athletes (OR, 3.00; 95% CI, 1.26-7.12; P = .01) and NC athletes (OR, 1.66; 95% CI, 1.08-2.56; P = .02). A similar relationship was noted across the entire study interval as well (MC vs SC: OR, 3.00; 95% CI, 1.26-7.12; P = .01; and MC vs NC: OR, 1.66; 95% CI, 1.07-2.56; P = .02) (Table 3).

Last, the time to lower extremity injury after RTP from concussion was significantly shorter in the MC group (P = .01), although there was no significant difference between the SC and NC groups. The difference in injury-free survival by log rank test was observed at 1 year after RTP (P = .007) but not at 90 days (P = .21) (Figure 2).

DISCUSSION

To our knowledge, the present study is the first to demonstrate a potential relationship between MC and an increased rate of lower extremity injury at time intervals long after the initial head trauma. Interestingly, athletes sustaining 1 concussion did not differ significantly from matched healthy controls with respect to lower extremity injury at any point during this study. However, athletes with MC were found to have a significantly greater odds of lower extremity injury and shorter time to lower extremity injury than matched SC and NC controls over the course of their career. This relationship held true when examining a previously reported study interval of 1 year after return to play, although it failed to reach significance at 90 days. Time to lower extremity injury was significantly shorter in the MC group, although no detectable difference existed between the SC and NC athletes. This suggests that some athletes that may be predisposed to higher rates of lower extremity injury after concussion or may be more injury prone due to player behavior.

Unique to this study, athletes with multiple career concussions were found to have a significantly greater odds of having sustained a lower extremity injury than athletes with 1 or no previous concussions. Current literature suggests that MC pose significant consequences in the form of increased risk for depression, cognitive deficits, and neurodegenerative disease.[9,24,25] However, fewer studies have

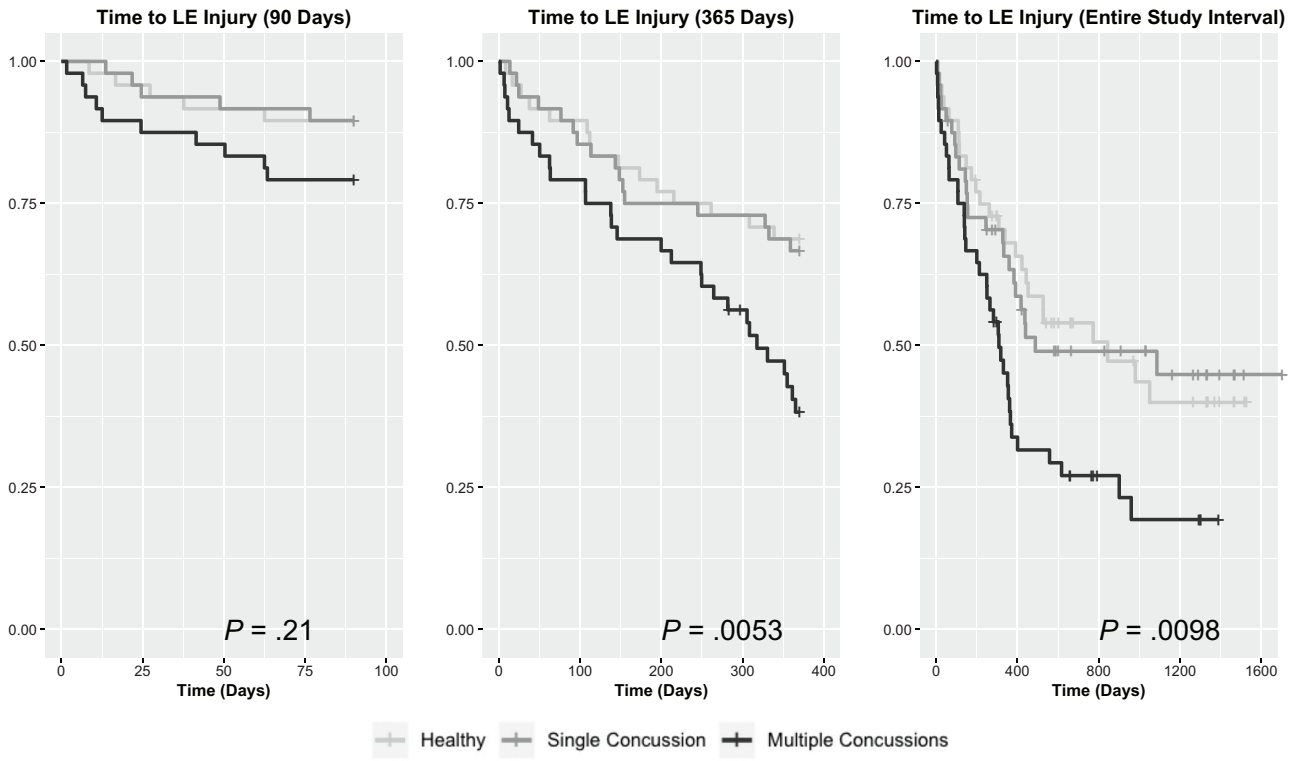

Figure 2. Time to lower extremity (LE) injury.

examined how MC may affect future injury risk. Guskiewicz et al[17] identified a cohort in whom concussion increased the risk of future concussion. In that study, athletes with a previous concussion were 1.4 times more likely to sustain a repeat concussion during the 3-year study interval. Furthermore, athletes with a history of 2 or 3 previous concussions exhibited a stepwise increase in risk. Guskiewicz et al hypothesized that a concussion may place the brain in a state of "neuronal vulnerability," leading to increased risk of injury and slower recovery. Such findings support the notion that athletes with MC may experience neurologic changes that increase a player's chance for future lower extremity injury. This hypothesis seems increasingly plausible when considering the outcomes from studies using dual-task gait testing or sophisticated neuroimaging and proprioceptive examination after concussion. In these instances, concussed athletes were found to have measurable differences in neuromuscular control that could factor into increased musculoskeletal injury risk.[19,20] Despite this, the precise mechanisms by which repeated head trauma affects normal physiological characteristics are debated and require more controlled experimentation. Irrespective, all athletes sustaining a concussion may benefit from more strict rehabilitation and RTP protocols to prevent reinjury.

Deficits in neuromuscular control after a concussion may not entirely explain the increase in odds of lower extremity injury, given that we noted a significant difference in outcomes between MC and SC athletes and no difference between SC and NC athletes. Although this study may have been underpowered to appreciate a measurable effect, these findings could also indicate that only a subset of concussed athletes are placed at increased risk for future injury and/or there is some other confounding factor between MC and lower extremity injury. One possible explanation may be the inherent differences in player behavior. More aggressive play could result in increased odds of all injuries and may explain why athletes with MC also have more lower extremity injuries. A study by Burman et al[7] seems to support this notion, as they found that athletes had elevated rates of injury both before and after documented concussion. Conversely, player behavior could also be adjusted as a result of a concussive event to avoid future injury and may explain why the SC athletes in the current study were less injury prone. To further evaluate this, we performed secondary analysis to determine the association between previous lower extremity injury history and number of sustained concussions. Interestingly, no significant differences were found among the 3 groups, suggesting further that injury-prone behavior may play a lesser role (Appendix Table A1, available in the online version of this article).

Player behavior or residual deficits in neuromuscular control could also explain why the location of lower extremity injury differs in athletes with MC. Interestingly, in this study, MC athletes sustained significantly more injuries to the foot. Secondary analysis revealed no difference in nature or location of injury. Given current limitations in the understanding of the pathophysiological process of

concussion, there is no clear explanation for why this injury profile differs. It is possible that proprioceptive deficits after concussion largely manifest distally, thus placing athletes at increased risk for more injuries to the foot. Lu et al[22] noted that hemispheric brain injury leads to significantly impaired proprioception of the contralateral limb, with the most profound deficits observed distally. This, in turn, could explain the measurable changes an athlete experiences in gait, balance, and leg stiffness after concussion.[12,13,16] This mechanism could also explain why other studies have found increased risk of more proximal lower extremity injuries after concussion. Namely, given the biomechanics of the lower extremity, proper foot proprioception is crucial for mitigating injury to other segments of the lower extremity. For example, loading of the lower extremity with toe-outward positioning of the foot leads to increased valgus stress on the knee, increasing risk for anterior cruciate ligament injuries.[26] Thus, an argument can be made to integrate sensitive pre- and postconcussion distal proprioceptive testing in sport. Doing so may allow the clinician to accurately measure a concussed athlete's return to his or her proprioceptive baseline during recovery.

Time to lower extremity injury was also significantly shorter in the MC cohort compared with the SC and NC cohorts. In contrast with studies by Brooks et al[6] and Herman et al,[18] the current study did not find any significant differences within 90 days of RTP. Rather, differences could be seen 1 year after RTP and across the entire study interval.[21,23] This finding is likely due to a combination of low sample size and study design. Specifically, at the 90-day RTP interval after concussion, many of the athletes in the MC group had yet to experience a second concussion. As such, there is no substantial difference in exposure at the early time points of this study. Furthermore, if athletes did experience a second concussion in this 90-day window, time spent recovering would decrease their exposure to potential lower extremity injuries relative to the SC athlete (Appendix Table A2 and Figure A1, available online). In short, these data are likely more meaningful as the duration of study increases. However, given that the results are significant as early as 1 year after the first documented concussion, this study may still provide useful clinical application. Specifically, these data suggest that MC could serve as an indicator that an athlete requires strict monitoring and rehabilitation to prevent subsequent lower extremity injury. Further study should aim to evaluate the differences between the MC and SC athlete to better understand what interventions can be implemented to prevent reinjury.

We considered the possibility that previous lower extremity injury history would influence time to injury outcomes. Secondary analysis via a Cox proportional hazards model demonstrated that sustaining a previous lower extremity injury was significantly associated with shorter time to injury across the entire study interval, although the number of concussions remained a significant factor as well. This analysis suggests that time to injury in these circumstances is likely multifactorial but still associated with repeated head trauma irrespective of prior lower extremity injury (Appendix Table A3, available online).

This study was not without limitations. As mentioned previously, many of the findings in this study regarding SC athletes and lower extremity injury risk in the imminent 90-day and 1-year intervals differed from findings in previous reports. This is likely due to the relatively small sample size used in this study. Further, after exclusions, this study was entirely represented by male athletes at a single athletic institution. As such, the conclusions in this study are likely not generalizable to female athletes and results may differ at other athletic centers. Similarly, most of the athletes under study were football players, and postconcussion injury patterns may differ in other sports. Last, as with all nonprospective studies, demographic and injury information could be subject to clerical error and variability based on operator input. Despite these limitations, this study has significant strengths. Given the relative infrequency with which serial concussions occur within the confines of a collegiate career, this study provides one of the largest examinations of MC in sport to date. In addition, this is the only study to explore the relationship between MC and associated lower extremity injury risk. Finally, this study uses a large study period, extending beyond previously reported 90-day and 1-year intervals, providing potential support for persistence of concussion-related sequelae long after documented RTP.

CONCLUSION

This study suggests that athletes with multiple concussions are at greater risk for future lower extremity injury than athletes with 1 or no concussions. In addition, the multiply concussed athletes experience a shorter time to lower extremity injury than their peers. Taken with findings from prior studies, these results suggest that the risk of future lower extremity injury may compound with each incidence of head trauma and may be due in part to persistent neuromuscular control deficits or differences in player behavior. In turn, the number of concussions may serve as a useful prognostic indicator for future lower extremity injury and identifies a cohort warranting further investigation. Irrespective, the recent literature on the association between injury risk and concussion invites the need for improvements to modern concussion return to play protocols.

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